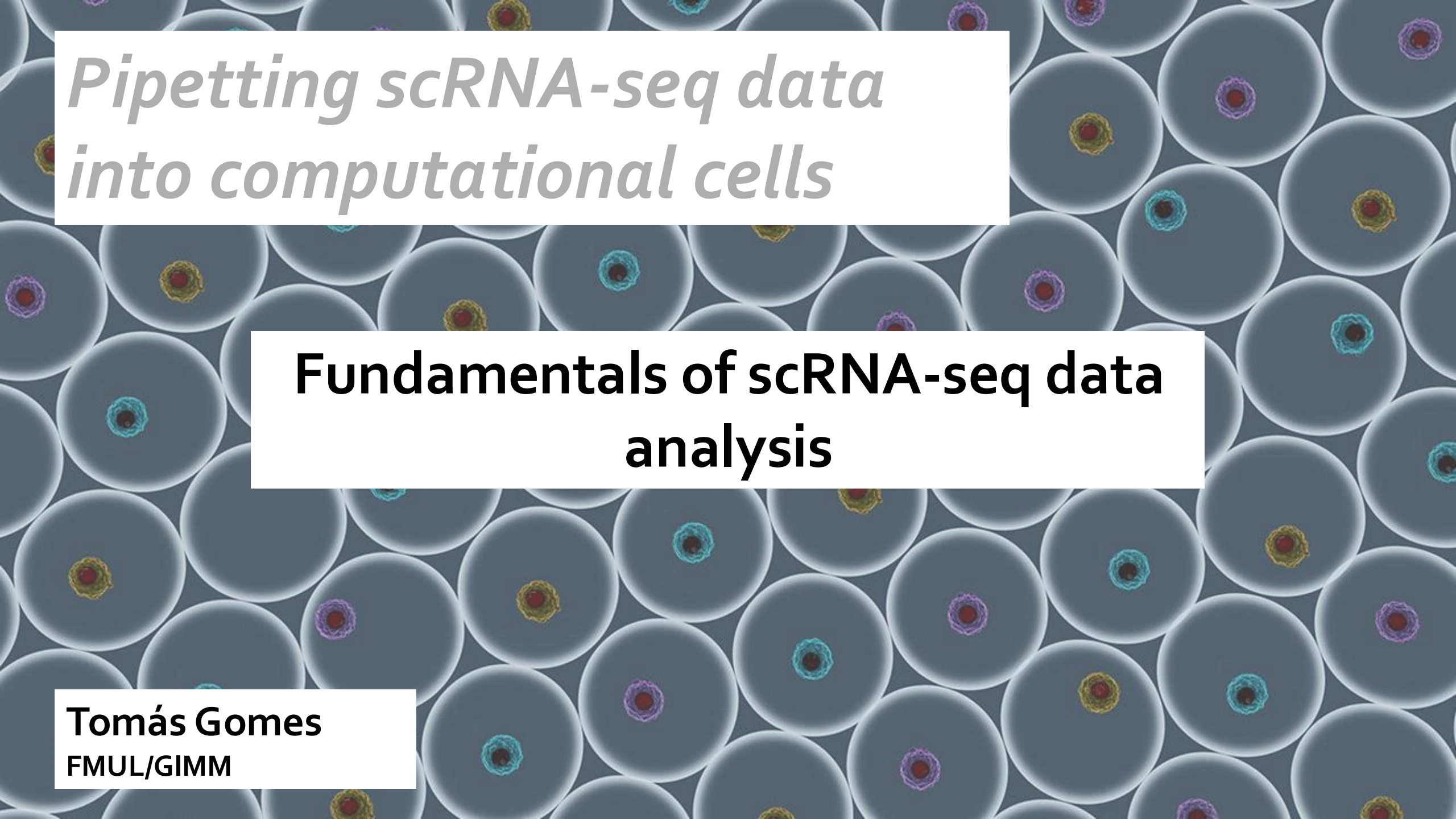


The background of the slide is a repeating pattern of stylized cells. Each cell is represented by a light blue circle with a darker blue outline. Inside each circle is a smaller, multi-colored shape representing a nucleus or organelle, with colors including red, yellow, green, and purple. The cells are arranged in a grid-like fashion, overlapping slightly.

*Pipetting scRNA-seq data
into computational cells*

Fundamentals of scRNA-seq data analysis

Tomás Gomes
FMUL/GIMM

Lesson topics

- The emergence of single-cell sequencing
- Processing and analysis steps
- Data integration and multi-modal data
- Uses of scRNA-seq in research



Lesson topics

- The emergence of single-cell sequencing
- Processing and analysis steps
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- Uses of scRNA-seq in research



Why transcriptomics?

- Able to capture **all transcripts** with little to no bias
- **Good proxy** for understanding the phenotype of a cell/tissue
- Easy to **access**
- Easy to **amplify the signal**



Bulk vs single-cell sequencing



Bulk vs single-cell sequencing

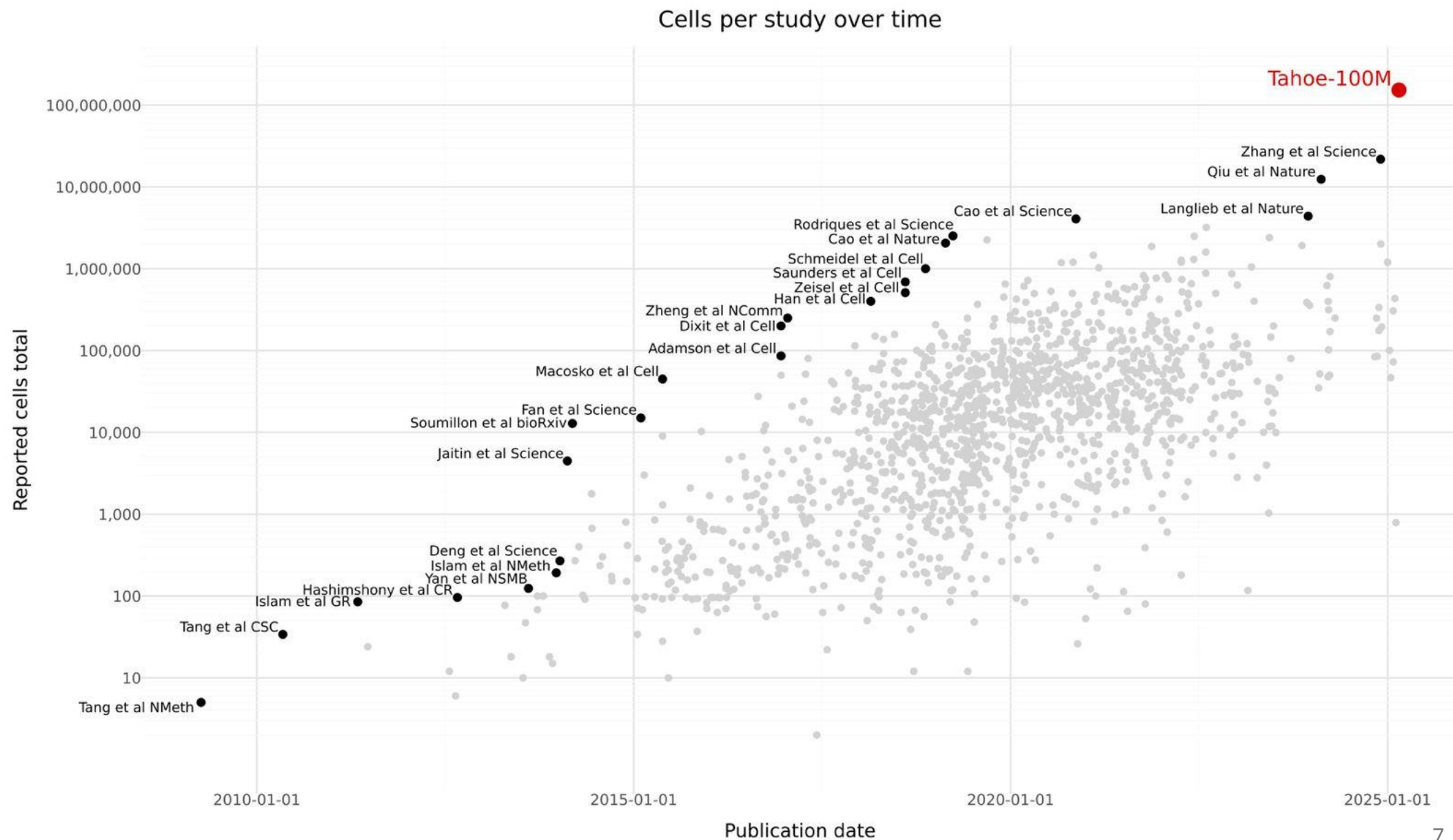
Bulk sequencing	Single-cell sequencing
cheaper	More expensive
Simpler protocol	More steps/more sensitive
Lower resolution	Higher resolution



This implies a cell-centric reasoning, i.e. at life's unit level.



The single-cell decade(s)



Setting the stage for the single-cell revolution

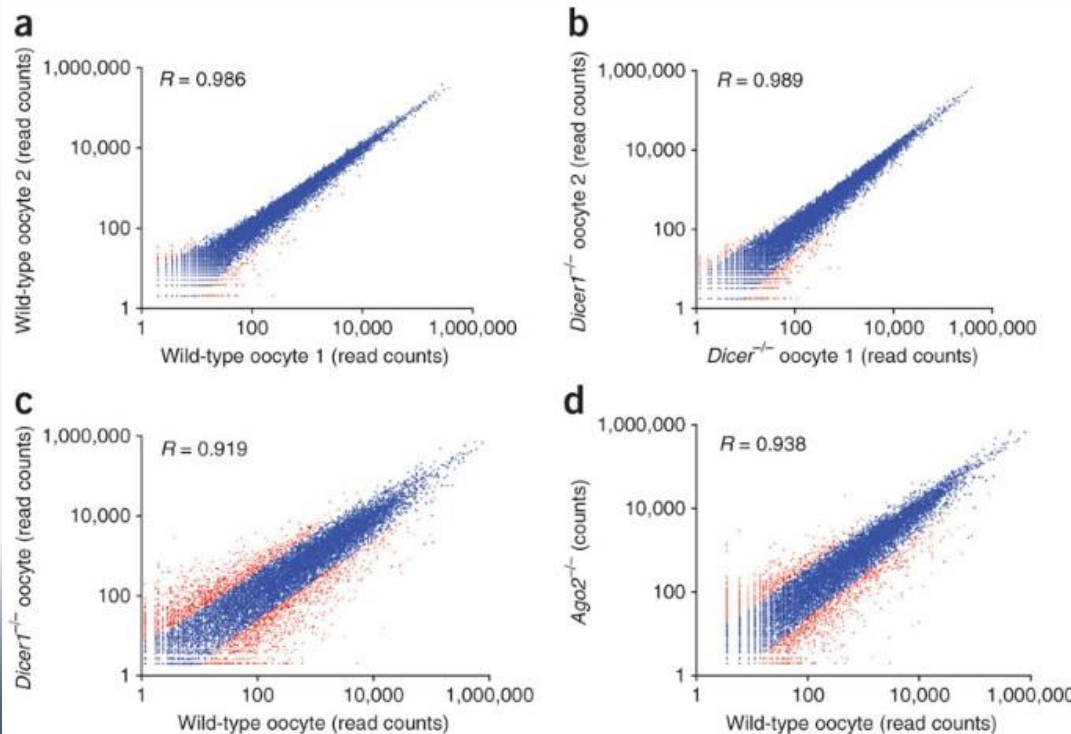
Published: 06 April 2009

mRNA-Seq whole-transcriptome analysis of a single cell

Fuchou Tang, Catalin Barbacioru, Yangzhou Wang, Ellen Nordman, Clarence Lee, Nanlan Xu, Xiaohui Wang, John Bodeau, Brian B Tuch, Asim Siddiqui, Kaiqin Lao ✉ & M Azim Surani ✉

Nature Methods 6, 377–382 (2009) | [Cite this article](#)

14k Accesses | 1308 Citations | 89 Altmetric | [Metrics](#)



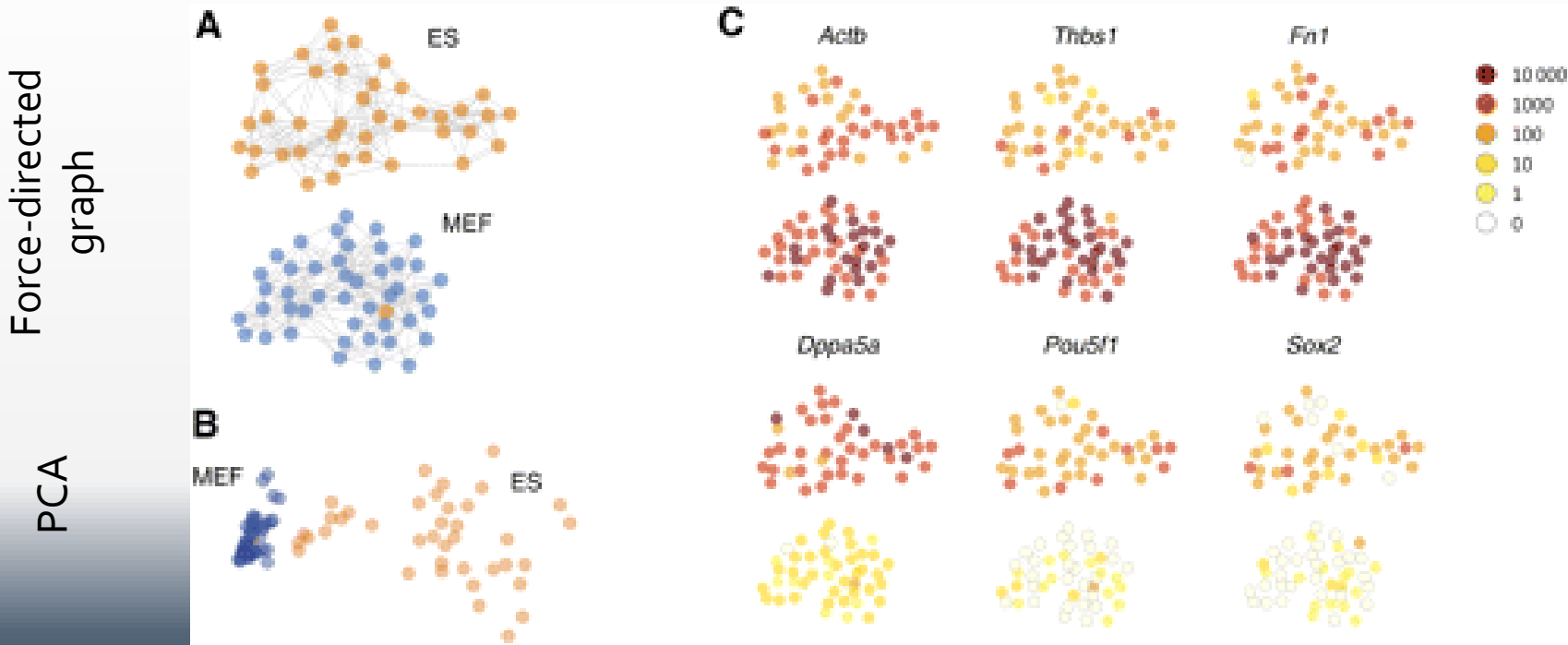
Goal was to **quantify RNA in a very low input system** (but with many different genes expressed)

Just 8 cells

2009 – just 1 year after the first RNA-seq paper

Setting the stage for the single-cell revolution

We envisage the future use of very large-scale single-cell transcriptional profiling to build a detailed map of naturally occurring cell types, which would give unprecedented access to the genetic machinery active in each type of cell at each stage of development.



The goal of using scRNA-seq is...

To reveal cellular heterogeneity
associated with gene expression



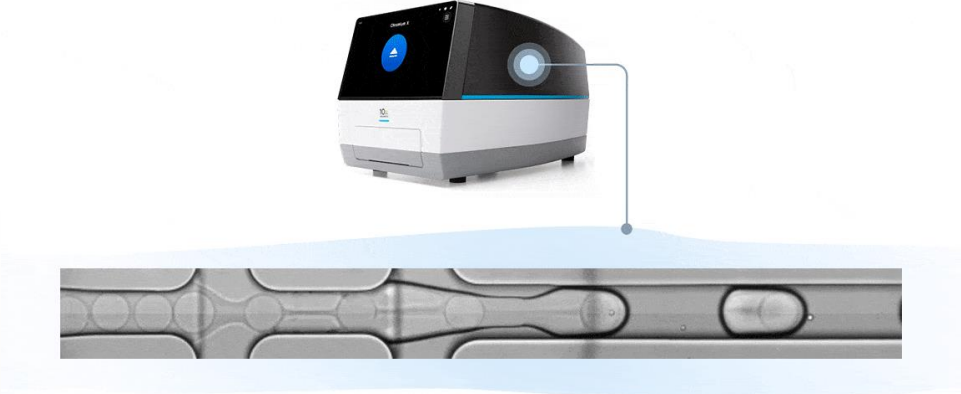
Lesson topics

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- Processing and analysis steps
- Data integration and multi-modal data
- Uses of scRNA-seq in research



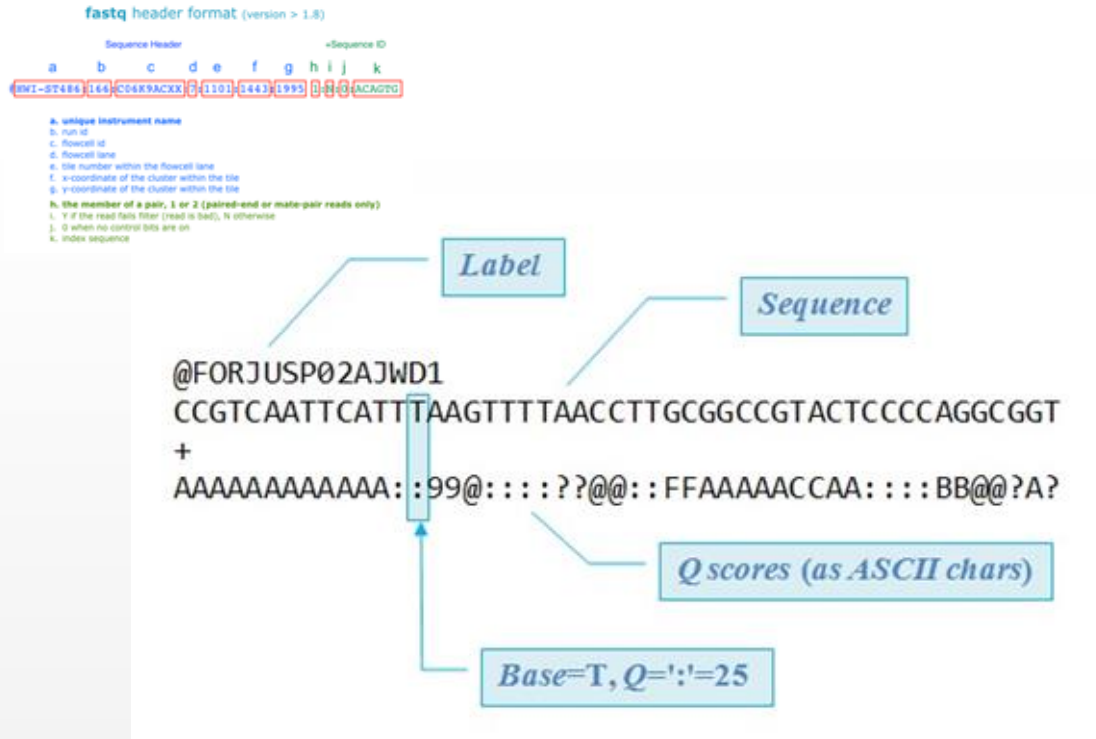
Common steps in a scRNA-seq experiment

scRNA-seq experiment	Bench work	Cell isolation
		Library construction
		Sequencing
	Computational work	Quantification
		Quality control
		Normalisation
		Dimensionality reduction
		Clustering/Trajectory analysis
		Differential expression
		...



File formats

FASTQ – (decoded) sequencing output



Count matrix

		Cells						
Genes	0	0	1	2	1	0	0	0
	0	1	3	5	2	0	1	0
	0	0	0	1	1	0	0	1
	3	2	0	0	1	0	0	0
	2	2	0	0	0	0	0	0
	1	1	1	0	0	3	1	2
	1	2	0	0	0	1	1	1
	1	2	0	0	0	2	2	2
	2	1	0	0	0	4	4	3
	0	0	0	0	0	2	1	1
	1	0	0	0	0	2	3	1
	0	0	0	0	0	2	1	1

What is sequenced?

FASTQ – (decoded) sequencing output

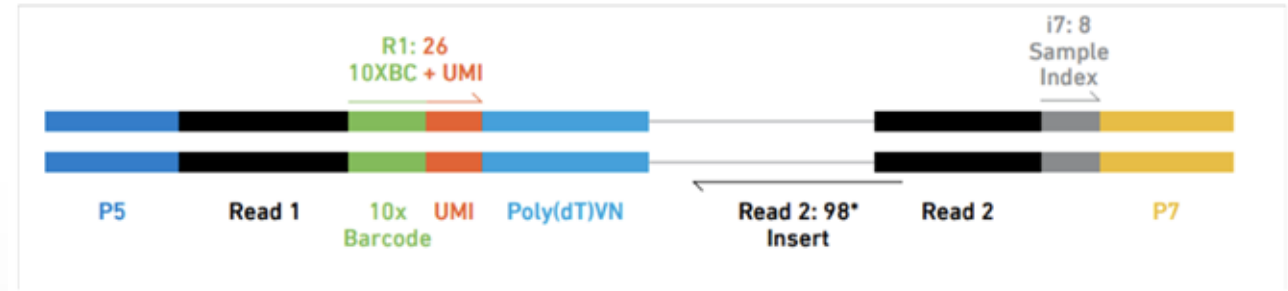
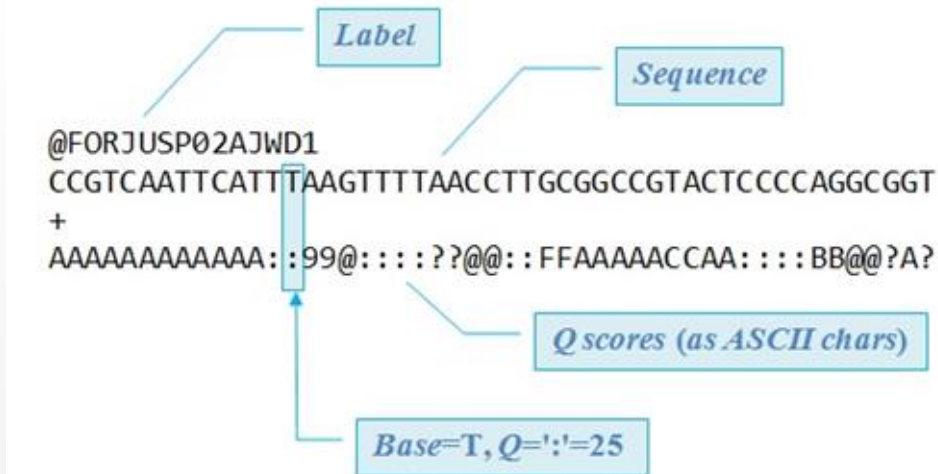


Fig. 2. Schematic of a fragment from a final Chromium™ Single Cell 3' v2 library. *Can be adjusted.

Number of fastq files received depends on method

In library construction, sample barcodes are added to split reads computationally (usually done as part of the sequencing service)

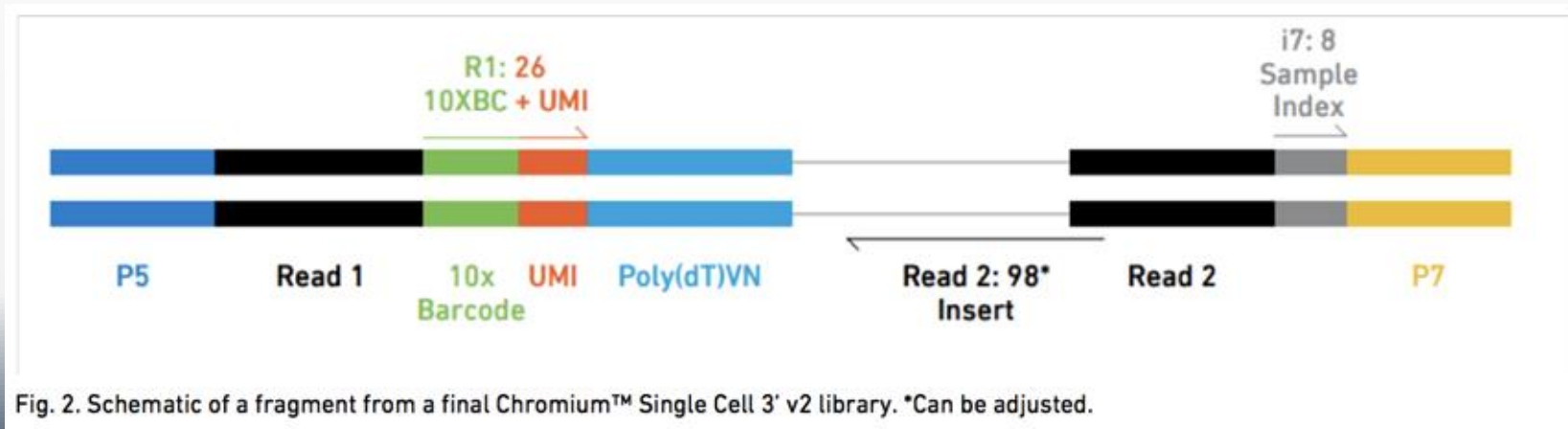
- For plate-based methods, each sample is a well, and therefore a cell
- For droplet-based methods, a sample is a mix of cells

What is sequenced?

In addition to sequencing the cDNA molecule, other artificial constructs are added:

- Illumina primer sequences
- **Cell barcode** – each cell is associated with a unique artificial sequence, which allows **grouping sequenced reads by the cell of origin**
- **Unique Molecular Identifier (UMI)** - To **avoid counting the same transcript twice** (usually caused by PCR duplicates), a sequence is added to each reverse transcribed RNA

This also means that counting UMI duplication can be a QC metric for the sample/sequencing



Quantification and demultiplexing - Cell Ranger

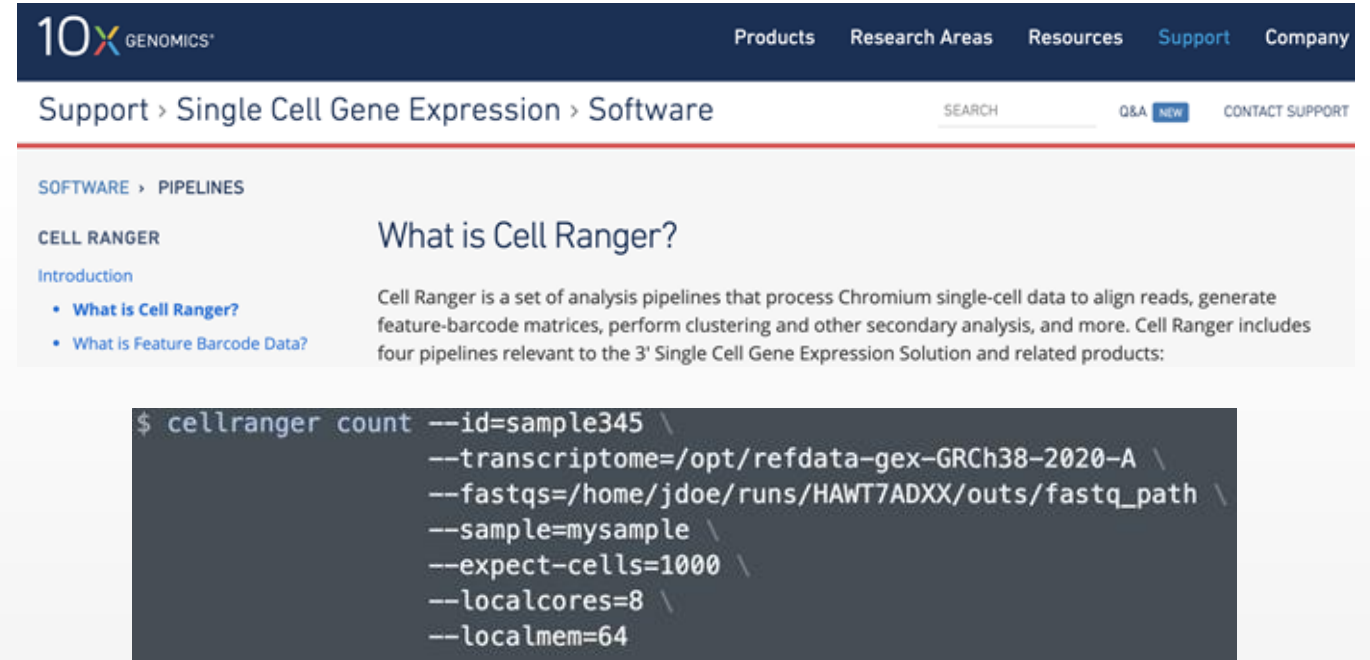
A command-line software suite to quantify (and analyse) data from 10x Genomics scRNA-seq/snRNA-seq

Based on STAR for alignment

Parses Cell barcodes and UMIs

Detects real cells and empty droplets

Performs some default analysis



The screenshot shows the 10x Genomics website's support page for Cell Ranger. The header includes the 10x Genomics logo and navigation links for Products, Research Areas, Resources, Support, and Company. The breadcrumb trail reads 'Support > Single Cell Gene Expression > Software'. The left sidebar lists 'SOFTWARE > PIPELINES' and 'CELL RANGER' with an 'Introduction' link and two bullet points: 'What is Cell Ranger?' and 'What is Feature Barcode Data?'. The main content area is titled 'What is Cell Ranger?' and contains a paragraph describing the software as a set of analysis pipelines for Chromium single-cell data. Below this is a terminal window showing the command to run 'cellranger count' with various options.

```
$ cellranger count --id=sample345 \
  --transcriptome=/opt/refdata-gex-GRCh38-2020-A \
  --fastqs=/home/jdoe/runs/HAWT7ADXX/outs/fastq_path \
  --sample=mysample \
  --expect-cells=1000 \
  --localcores=8 \
  --localmem=64
```

Running CellRanger

```
$ cellranger count --id=sample345 \  
  --transcriptome=/opt/refdata-gex-GRCh38-2020-A \  
  --fastqs=/home/jdoe/runs/HAWT7ADXX/outs/fastq_path \  
  --indices=SI-3A-A1
```

Martian Runtime - v4.0.8

Running preflight checks (please wait)...

2016-01-01 10:23:52 [runtime] (ready) ID.sample345.CELLRANGER_CS.CELLRANGER.SETUP_CHUNKS ...

2012-01-01 12:10:09 [runtime] (join_complete) ID.sample345.CELLRANGER_CS.CELLRANGER.SUMMARIZE_REPORTS

Outputs:

- Run summary HTML: /home/jdoe/runs/sample345/outs/web_summary.html
- Run summary CSV: /home/jdoe/runs/sample345/outs/metrics_summary.csv
- BAM: /home/jdoe/runs/sample345/outs/possorted_genome_bam.bam
- BAM index: /home/jdoe/runs/sample345/outs/possorted_genome_bam.bam.bai
- Filtered feature-barcode matrices MEX: /home/jdoe/runs/sample345/outs/filtered_feature_bc_matrix
- Filtered feature-barcode matrices HDF5: /home/jdoe/runs/sample345/outs/filtered_feature_bc_matrix.h5
- Unfiltered feature-barcode matrices MEX: /home/jdoe/runs/sample345/outs/raw_feature_bc_matrix
- Unfiltered feature-barcode matrices HDF5: /home/jdoe/runs/sample345/outs/raw_feature_bc_matrix_h5.h5
- Secondary analysis output CSV: /home/jdoe/runs/sample345/outs/analysis
- Per-molecule read information: /home/jdoe/runs/sample345/outs/molecule_info.h5
- Loupe Browser file: /home/jdoe/runs/sample345/outs/cloupe.cloupe

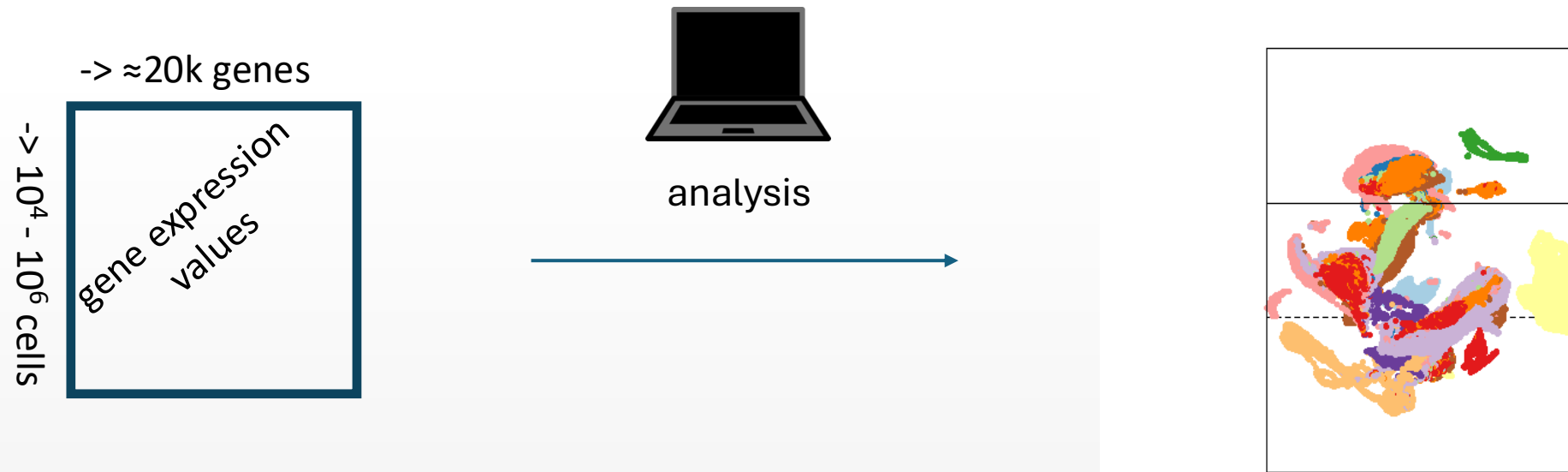
Pipestance completed successfully!

Activity – reading the CellRanger Web Summary

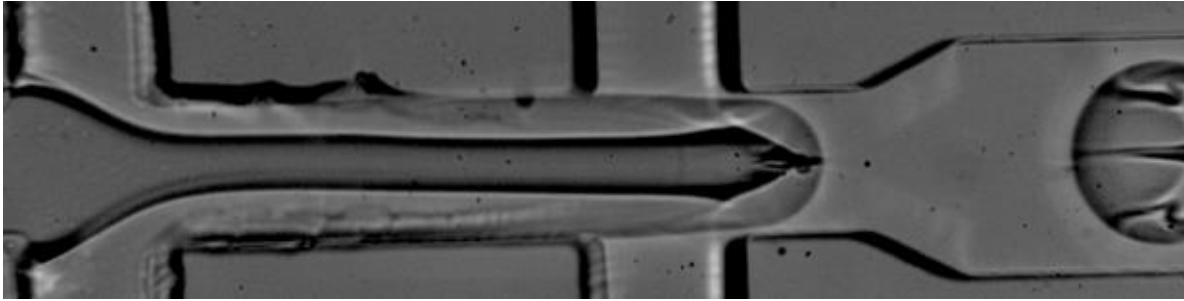
1. Obtain the web_summary.html from the course github
2. Explore the web_summary.html output file. Which are the most important metrics to determine whether your experiment was successful? Why?



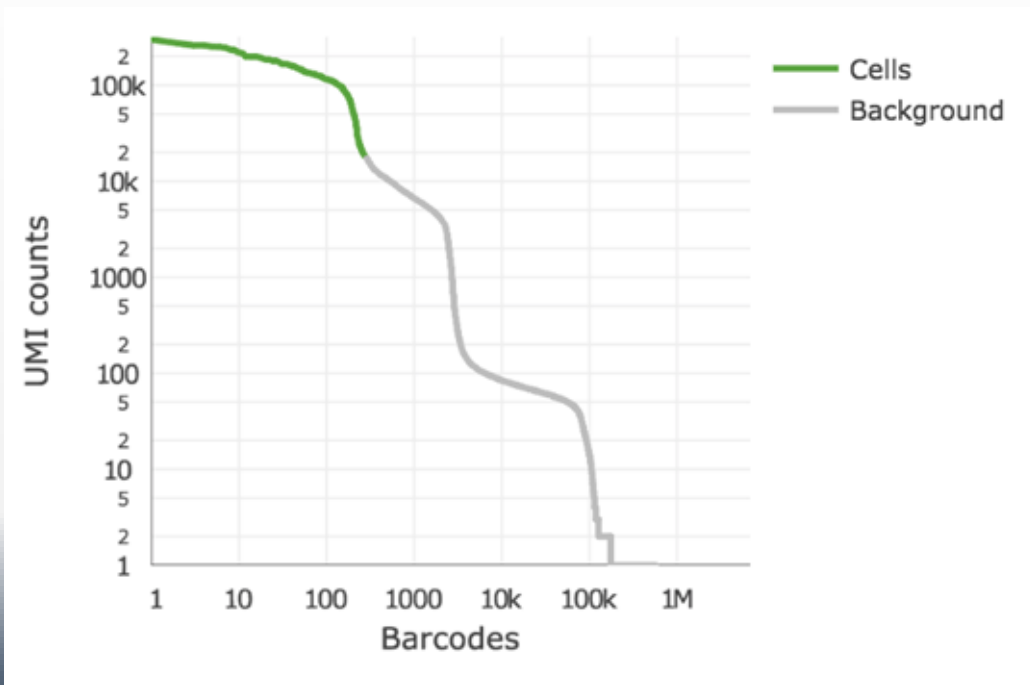
Analysing scRNA-seq data



What is in a microdroplet?



*this is from Drop-seq, but in the Chromium the principle is similar



Droplets can:

- contain nothing – **empty droplet**
- contain RNA but no cell – **empty with ambient RNA**
- contain a cell – **real cell**
- contain more than one cell - **doublet**

Droplets classified based on:

- UMI counts
- Transcriptomic similarity with empty droplets
- Transcriptomic similarity between cell populations

UMI counts and **# genes** are important QC metrics

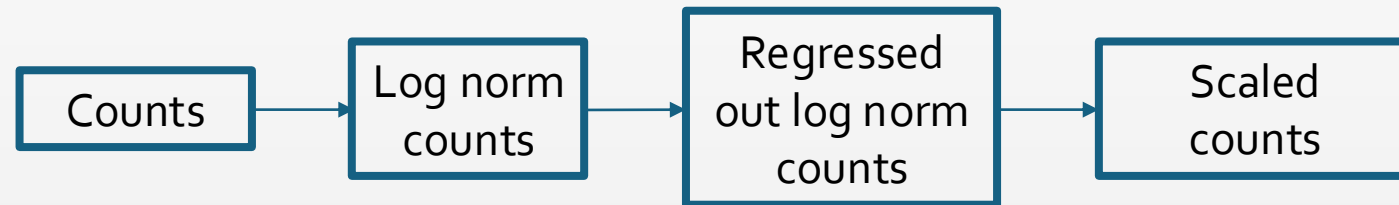
- High counts can indicate doublets
- Low counts can indicate dead cells/empty droplets

Specific gene sets are also important in QC

- **MT genes** – high in dying cells
- **Stress genes** (Heat Shock Proteins, ...) – tissue dissociation effects

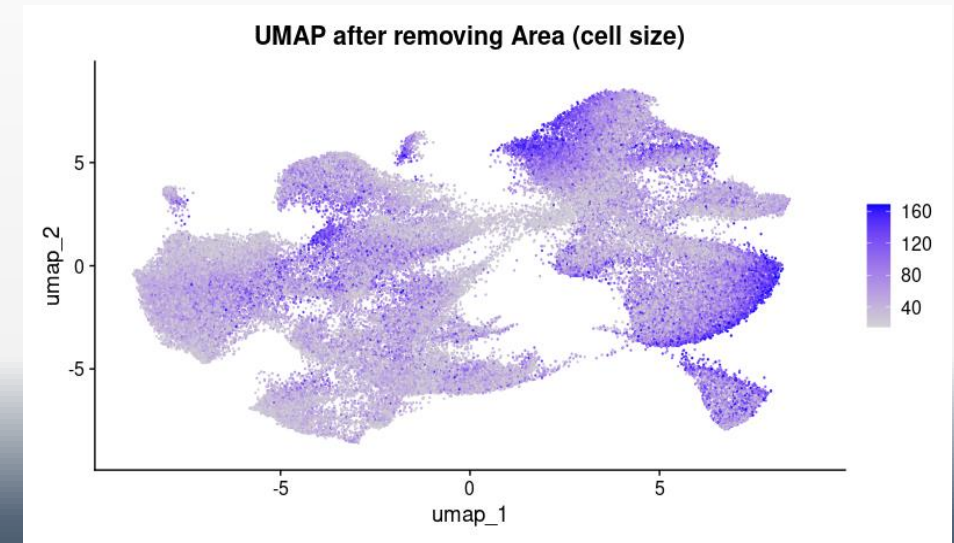
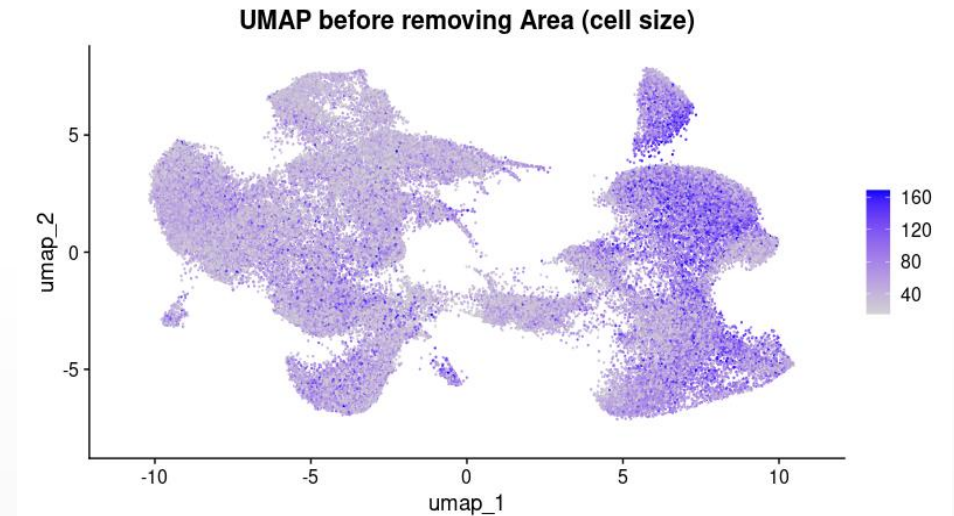
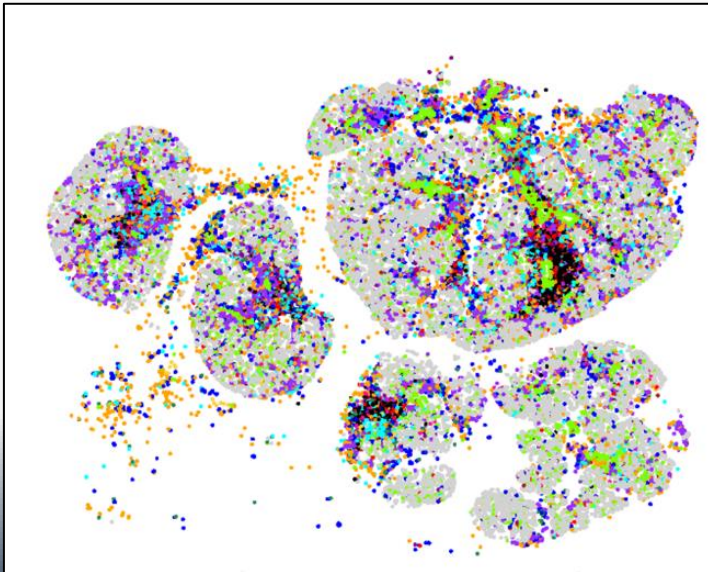
Importance of normalisation

- The most common way to normalise scRNA-seq data is **dividing UMI counts by the total per cell**, followed by a **log-transformation**
- A “pseudocount” has to be added to non-expressed genes (since you can’t have $\log(0)$) – usually 1, but ideally a fraction (0.1 or lower)
- In this step, **regressing out the effect of other technical/undesired variables** (e.g. total counts per cell, % MT genes, cell cycle) can also be performed
- Further **scaling** (i.e. subtract the mean (and optionally divide by the standard deviation)) ensures the data can be used for linear projections (e.g PCA)



Importance of correcting for technical variables

- **Regressing out variables** with a potential technical effect avoids grouping cells by similar technical effects

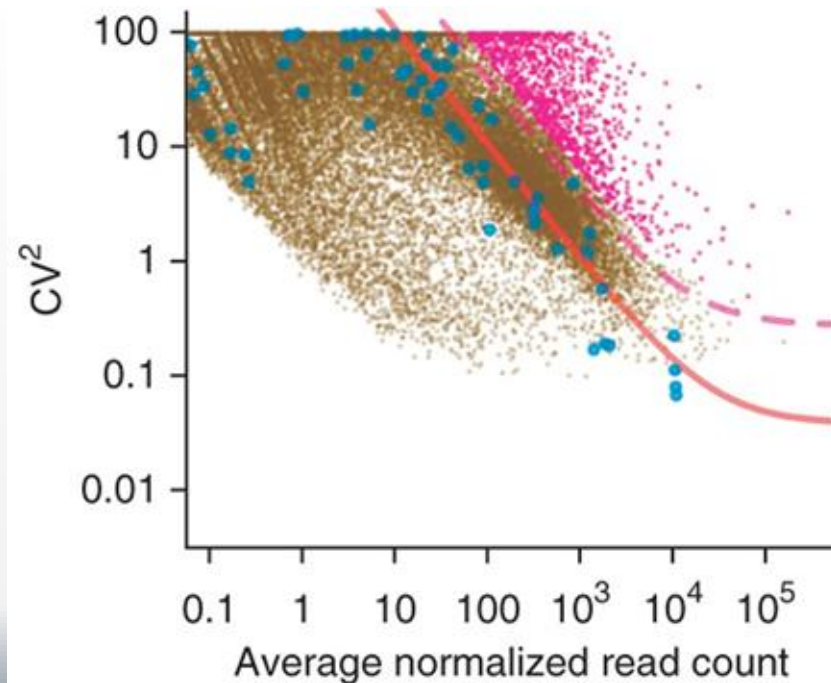


Obtaining highly variable (“meaningful”) genes

Published: 22 September 2013

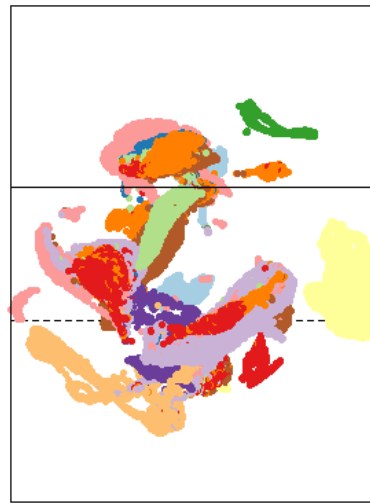
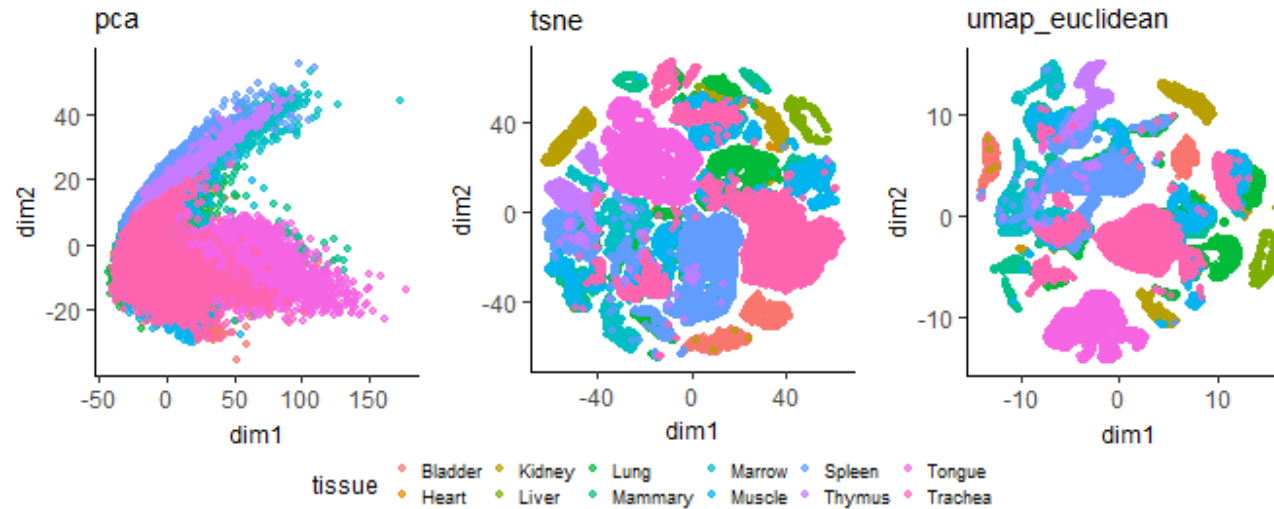
Accounting for technical noise in single-cell RNA-seq experiments

Philip Brennecke, Simon Anders, Jong Kyoung Kim, Aleksandra A Kołodziejczyk, Xiuwei Zhang, Valentina Proserpio, Bianka Baying, Vladimir Benes, Sarah A Teichmann, John C Marioni & Marcus G Heisler



- Genes don't all equally reflect biological variance. For example, housekeeping genes should be ubiquitously expressed, and therefore not as useful to distinguish cell types
- Additionally, reducing the feature (gene) space in scRNA-seq data helps speed up computation
- This gene subset should *only* be used for dimensionality reduction/clustering/trajectory analysis, not the characterization of cell populations (i.e. differential expression)
- Ultimately, there are various ways to obtain them, and those should be tuned to the dataset at hand

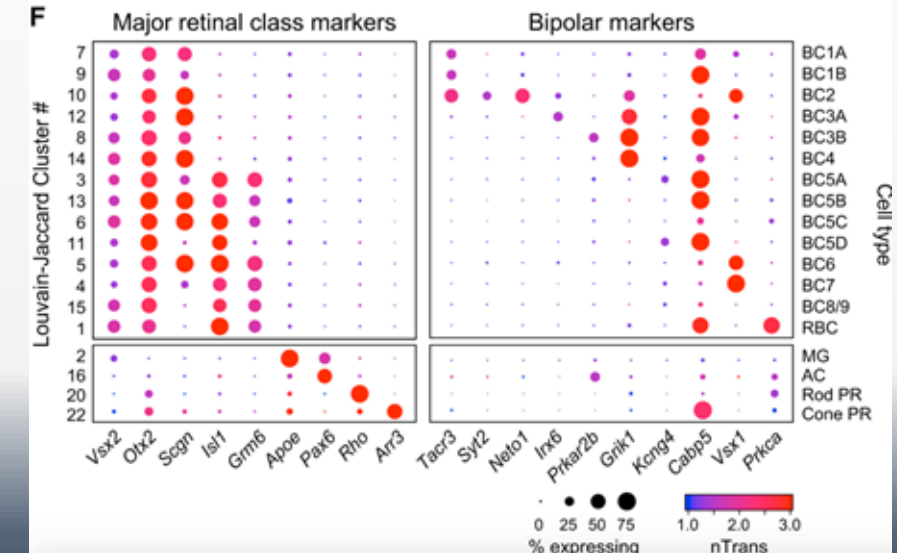
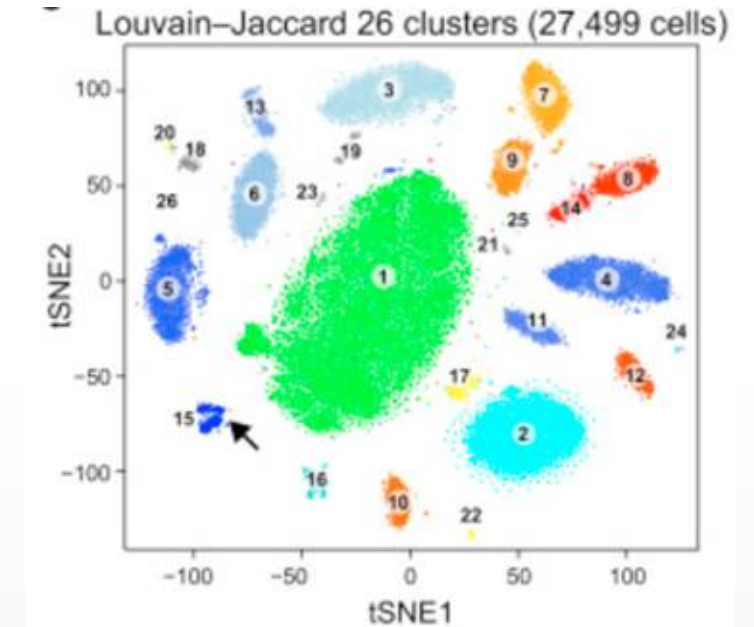
Dimensionality Reduction



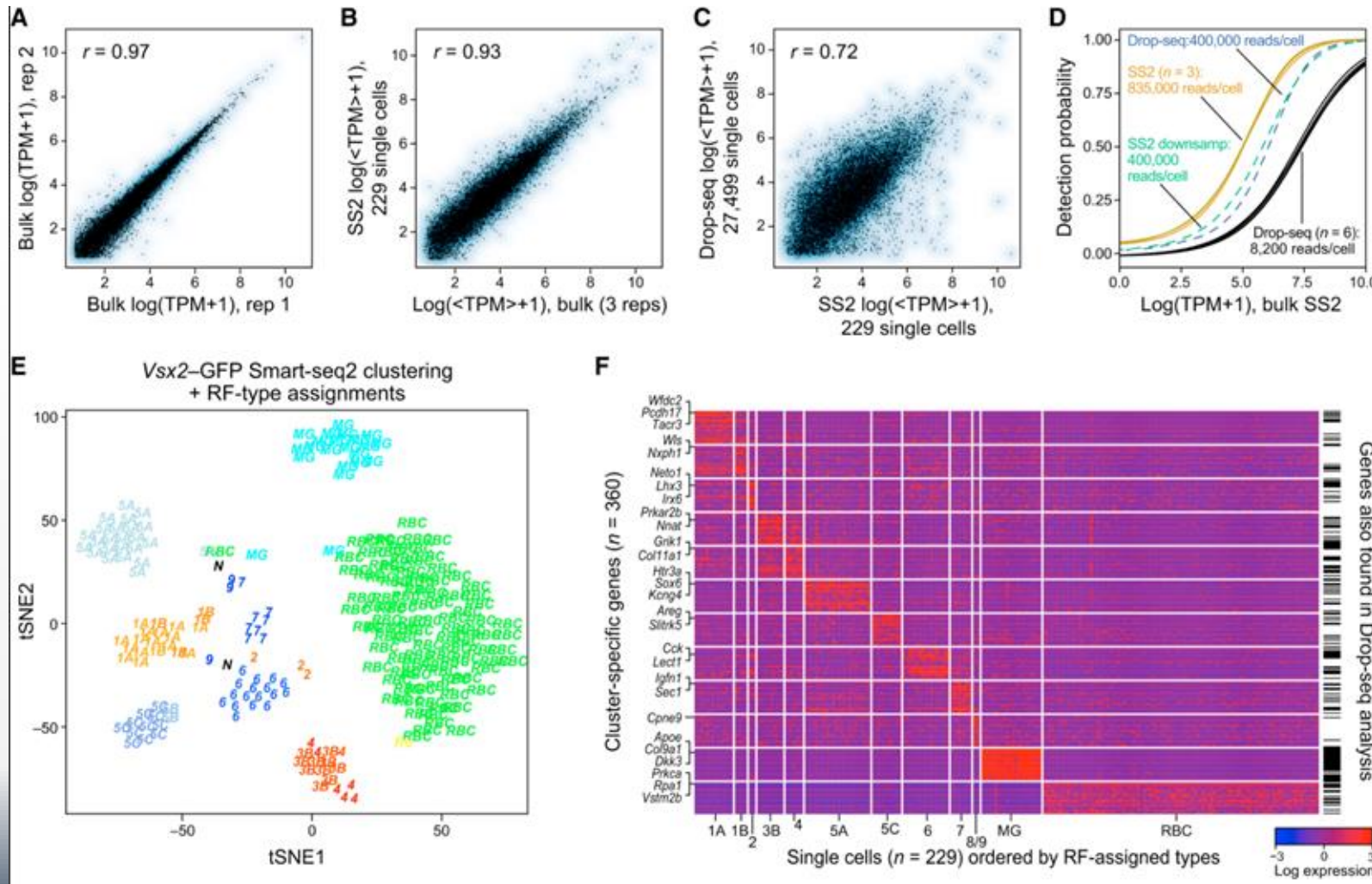
- scRNA-seq datasets commonly have ~10,000 *data points* (cells) and ~20,000 *features* (genes) – hard to visualise!
- **DR algorithms help us visualize the major variability** within a dataset
- PCA is the most common linear DR method, whereas tSNE and UMAP are non-linear
- PCA gives us orthogonal dimensions reflecting different aspects of the dataset's variability
- tSNE and UMAP are used to “flatten” the dataset into a visual 2D representation, **not to do any further analysis**, since they distort distances between cells

Identifying cell types - clustering

- Clustering is used to **identify discrete cell populations based on gene expression similarity**
- The most popular clustering methods, **Leiden** and **Louvain**, are **community detection methods** (clustering for graphs)
- They work on cell **neighbourhood graphs**, usually computed from **PCA space, NOT UMAP/tSNE** since these distort distances (similarity) between cells
- **Marker genes** are obtained with differential expression **testing of one population vs the rest**

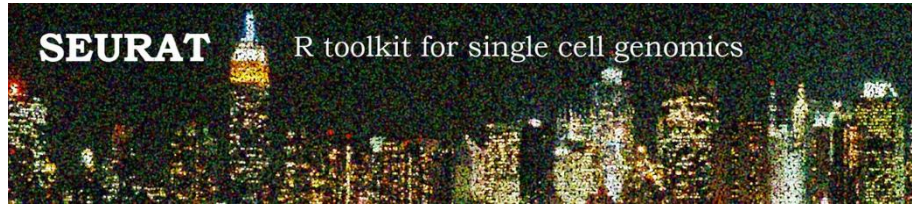


Identifying cell types from scRNA-seq



- **Plate** experiments tend to be sequenced deeper per cell – **more genes detected**
- **Droplet** experiments have less depth, but **capture more cells**
- To identify cell types, more cells is better than more genes. This is because not many genes are needed to define a cell type, but a good number of cells is necessary to find an independent cluster

Two major platforms for scRNA-seq analysis



```
library(Seurat)

# load data and create Seurat object
pbmc.data = Read10X(data.dir = "../data/pbmc3k/filtered_gene_bc_matrices/hg19/")
pbmc = CreateSeuratObject(counts = pbmc.data, min.cells = 3, min.features = 200)

# normalise data
pbmc = NormalizeData(pbmc, normalization.method = "LogNormalize", scale.factor = 10000)

# find highly variable genes
pbmc = FindVariableFeatures(pbmc, selection.method = "vst", nfeatures = 2000)

# scale data
pbmc = ScaleData(pbmc, features = rownames(pbmc))

# linear dimensionality reduction
pbmc = RunPCA(pbmc, features = VariableFeatures(object = pbmc))

# clustering
pbmc = FindNeighbors(pbmc, dims = 1:40)
pbmc = FindClusters(pbmc, resolution = 0.5)

# non-linear dimensionality reduction
pbmc = RunUMAP(pbmc, dims = 1:10)

# find marker genes
pbmc.mk = FindAllMarkers(pbmc, only.pos = TRUE, min.pct = 0.25, logfc.threshold = 0.25)
```

<https://satijalab.org/seurat/>



```
import scanpy as sc

# load data and create AnnData (scanpy) object
adata = sc.read_10x_mtx('data/filtered_gene_bc_matrices/hg19/',
var_names='gene_symbols')

# normalise data
sc.pp.normalize_total(adata, target_sum=1e4)
sc.pp.log1p(adata)

# find highly variable genes
sc.pp.highly_variable_genes(adata, min_mean=0.0125, max_mean=3,
min_disp=0.5)

# scale data
sc.pp.scale(adata, max_value=10)

# linear dimensionality reduction
sc.tl.pca(adata, svd_solver='arpack')

# clustering
sc.pp.neighbors(adata, n_neighbors=10, n_pcs=40)
sc.tl.leiden(adata)

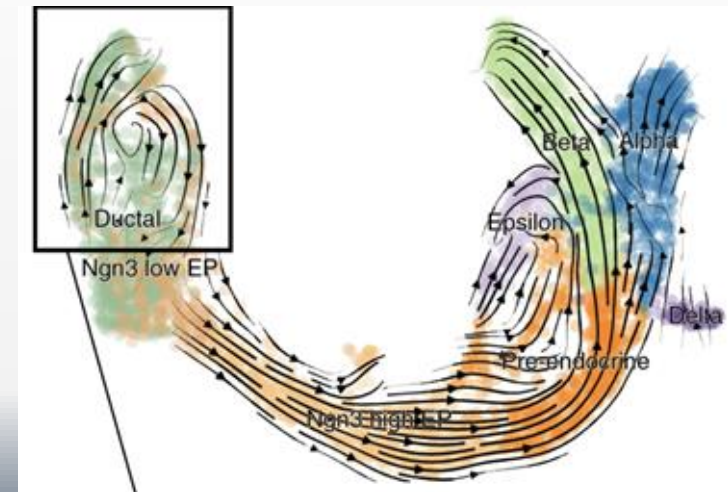
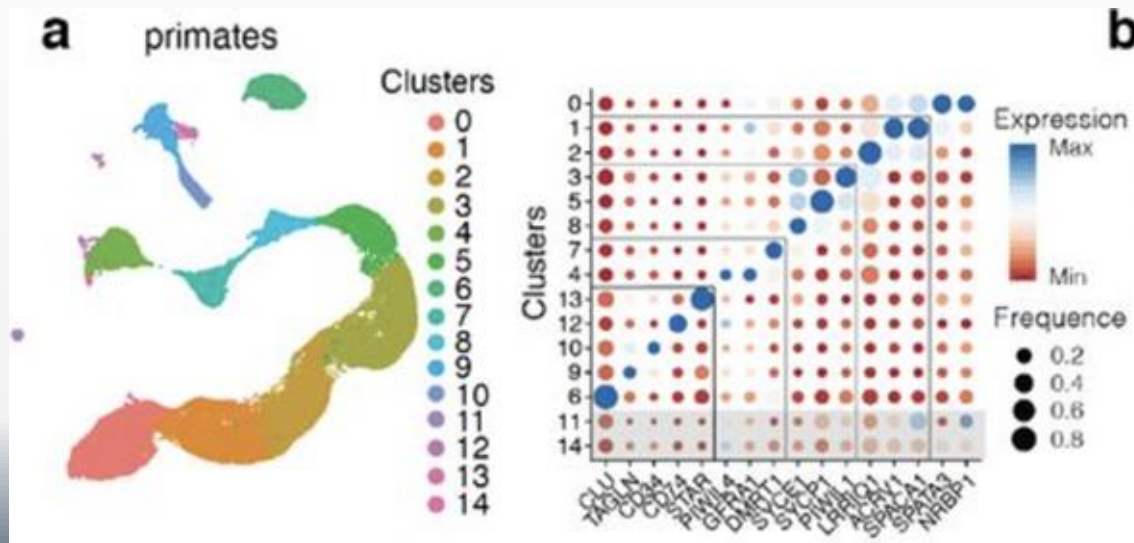
# non-linear dimensionality reduction
sc.tl.umap(adata)

# find marker genes
sc.tl.rank_genes_groups(adata, 'leiden', method='t-test')
```

<https://scanpy.readthedocs.io/en/stable/> <https://www.sc-best-practices.org/preamble.html>

Identifying trajectories from scRNA-seq

- scRNA-seq data can also identify **continuous** effects
- UMAP and tSNE can give an indication that a trajectory may exist, but should not be used to model it
- Pseudotime analysis allows for the identification of **continuously variable genes**
- **RNA velocity** can be inferred from spliced and unspliced reads to model cell changes



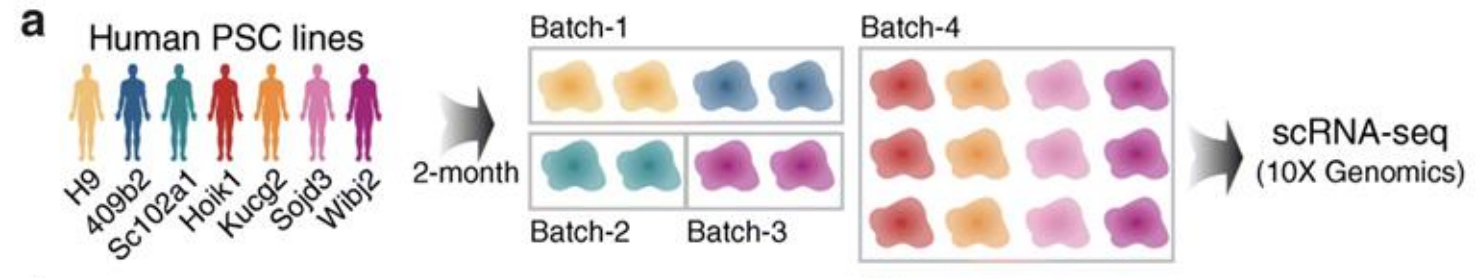
Lesson topics

- The emergence of single-cell sequencing
- Processing and analysis steps
- Data integration and multi-modal data
- Uses of scRNA-seq in research

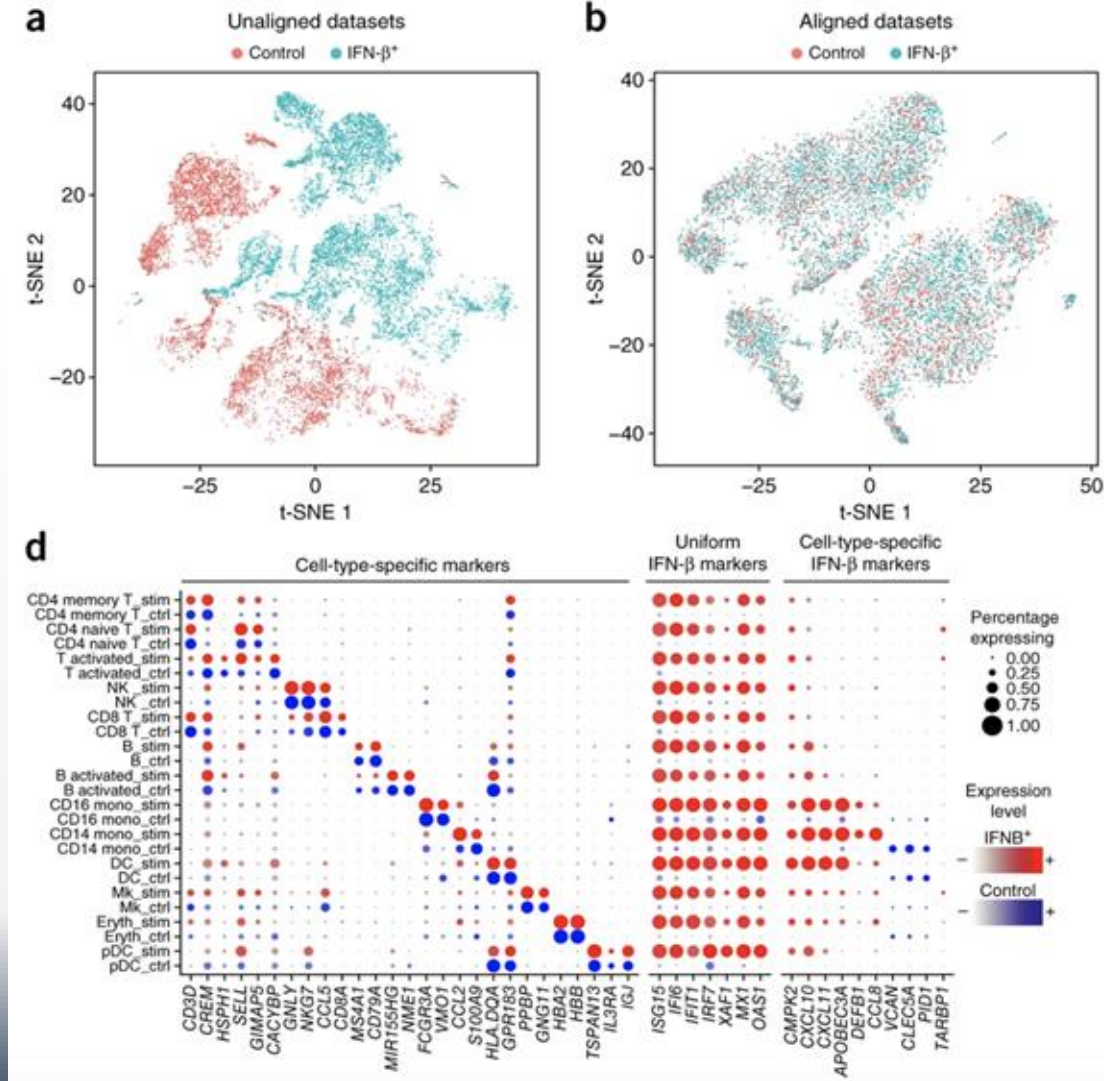


Data integration – correct unwanted variation

- Gene expression matrices reflect a variety of factors, both **technical** and **biological**
- The usual goal with scRNA-seq is to **represent cell types**
- Computational integration tools work on a low-dimension space (e.g. PCA) to **correct for unwanted technical variation** (e.g. donor, experiment day, isolation protocol, or lab)



Correcting unwanted (?) variation

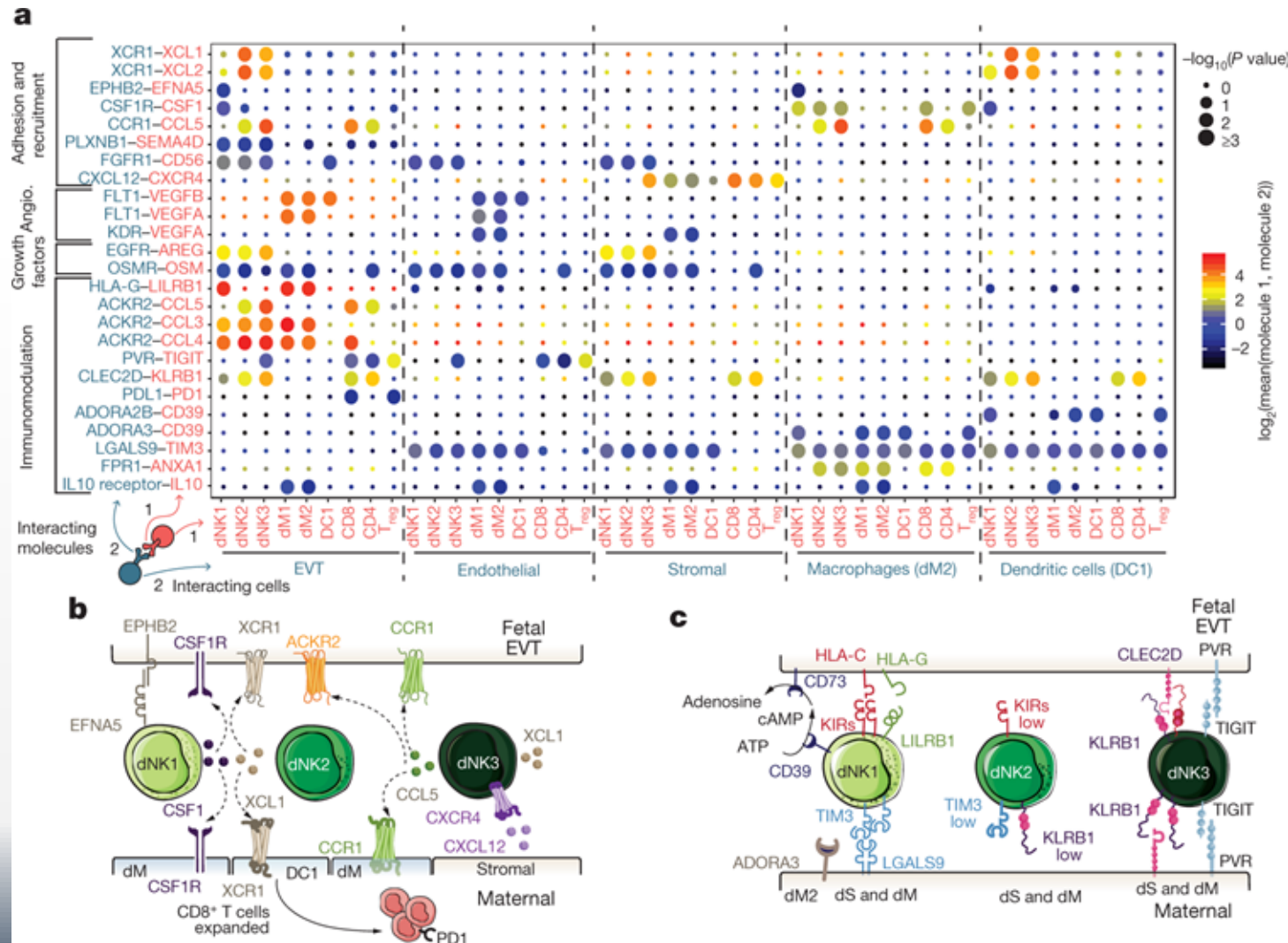


Cell types are harder to identify in multifactorial experiments, due to low biological variability

Similarly, some **lowly-represented cell types** only appear once the data is integrated (increase in cell sample numbers)

Common biological factors that are corrected for are tissue, treatment, or species

Extending gene expression interpretation



scRNA-seq can be used to infer cell-cell communication events

Lesson topics

- The emergence of single-cell sequencing
- Processing and analysis steps
- Data integration and multi-modal data
- Uses of scRNA-seq in research

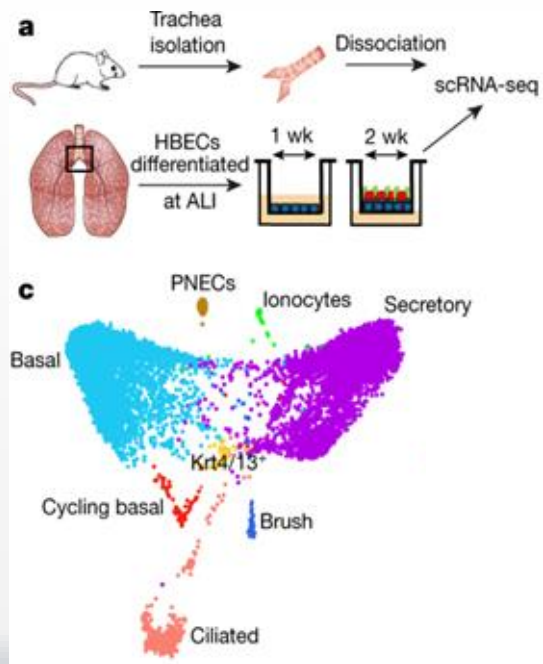


Uses of scRNA-seq – cell type discovery

Letter | Published: 01 August 2018

A single-cell atlas of the airway epithelium reveals the CFTR-rich pulmonary ionocyte

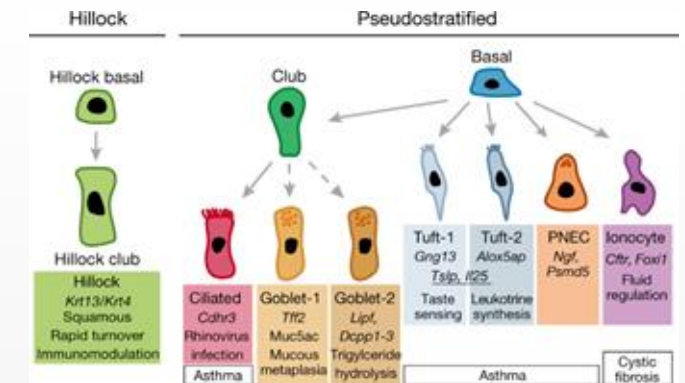
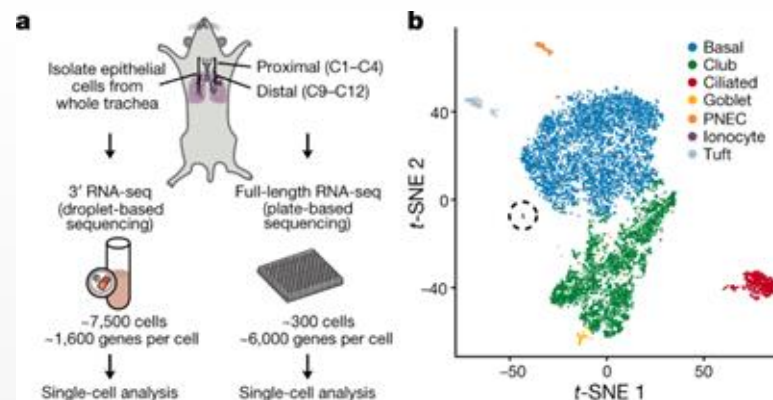
Lindsey W. Plasschaert, Rapolas Žilionis, Rayman Choo-Wing, Virginia Savova, Judith Knehr, Guglielmo Roma, Allon M. Klein & Aron B. Jaffe



Article | Published: 01 August 2018

A revised airway epithelial hierarchy includes CFTR-expressing ionocytes

Daniel T. Montoro, Adam L. Haber, Moshe Biton, Vladimir Vinarsky, Brian Lin, Susan E. Birket, Feng Yuan, Sijia Chen, Hui Min Leung, Jorge Villoria, Noga Rogel, Grace Burgin, Alexander M. Tsankov, Avinash Waghay, Michal Slyper, Julia Waldman, Lan Nguyen, Danielle Dionne, Orit Rozenblatt-Rosen, Purushothama Rao Tata, Hongmei Mou, Manjunatha Shivaraju, Hermann Bihler, Martin Mense, Guillermo J. Tearney, Steven M. Rowe, John F. Engelhardt, Aviv Regev & Jayaraj Rajagopal



Description of ionocytes in mouse (and human) airways

Specified by transcription factor Foxl1

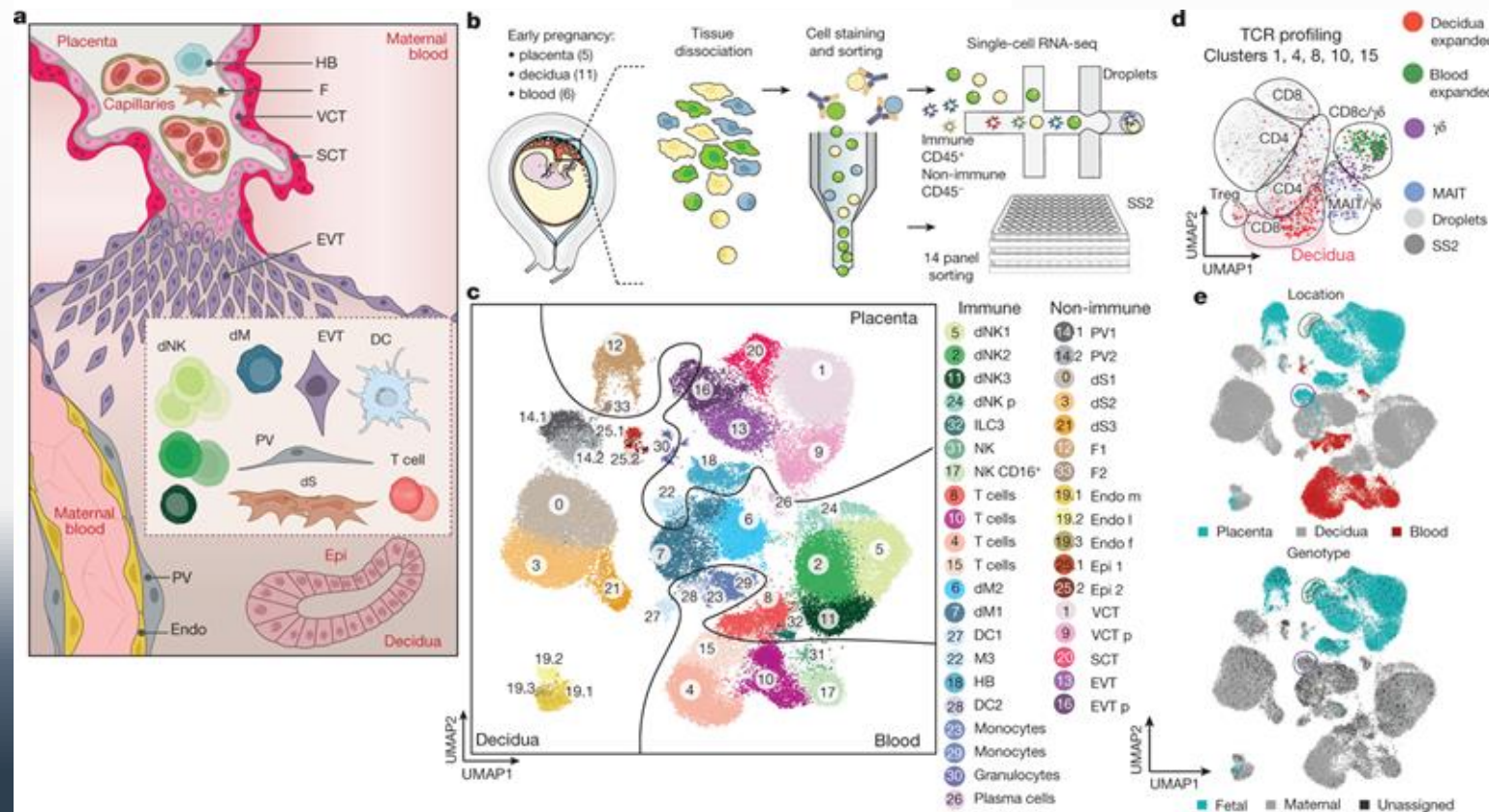
Involved in cystic fibrosis

Uses of scRNA-seq – tissue composition

Article | Published: 14 November 2018

Single-cell reconstruction of the early maternal–fetal interface in humans

Roser Vento-Tormo, Mirjana Efremova, Rachel A. Botting, Margherita Y. Turco, Miquel Vento-Tormo, Kerstin B. Meyer, Jong-Eun Park, Emily Stephenson, Krzysztof Polański, Angela Goncalves, Lucy Gardner, Staffan Holmqvist, Johan Henriksson, Angela Zou, Andrew M. Sharkey, Ben Millar, Barbara Innes, Laura Wood, Anna Wilbrey-Clark, Rebecca P. Payne, Martin A. Ivarsson, Steve Lisgo, Andrew Filby, David H. Rowitch, Judith N. Bulmer, Gavin J. Wright, Michael J. T. Stubbington, Muzlifah Haniffa ✉, Ashley Moffett ✉ & Sarah A. Teichmann ✉ -Show fewer authors



Identification of all cell populations at the maternal-fetal interface

Discovering cellular interactions in immune and non-immune compartment

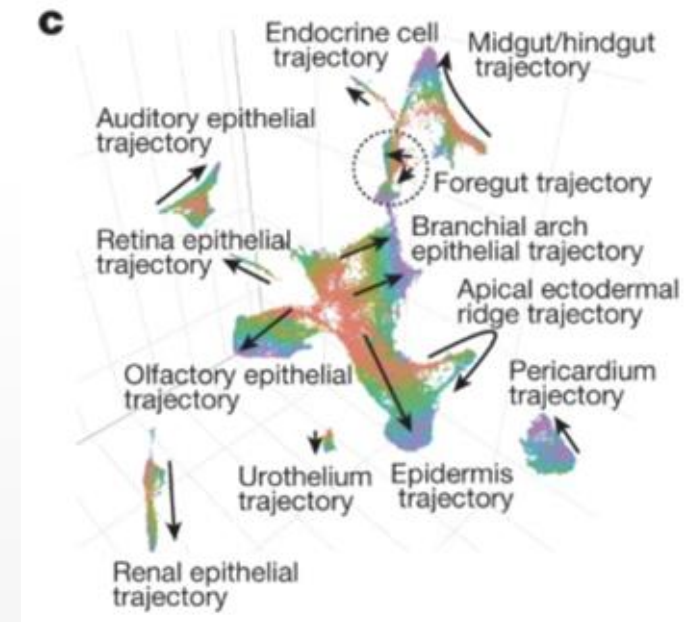
Unravel cell origin (maternal or fetal?) by combining variant calling with genome data

Uses of scRNA-seq – development

Article | Published: 20 February 2019

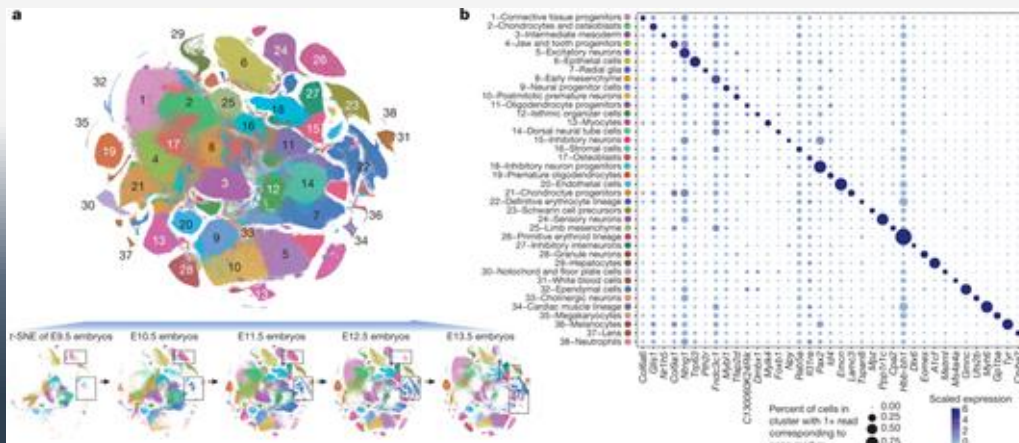
The single-cell transcriptional landscape of mammalian organogenesis

Junyue Cao, Malte Spielmann, Xiaojie Qiu, Xingfan Huang, Daniel M. Ibrahim, Andrew J. Hill, Fan Zhang, Stefan Mundlos, Lena Christiansen, Frank J. Steemers, Cole Trapnell & Jay Shendure



Profiling about 2M cells during mouse organogenesis

Defining the major developmental trajectories for cell types based on transcriptional changes



Uses of scRNA-seq – building gene networks

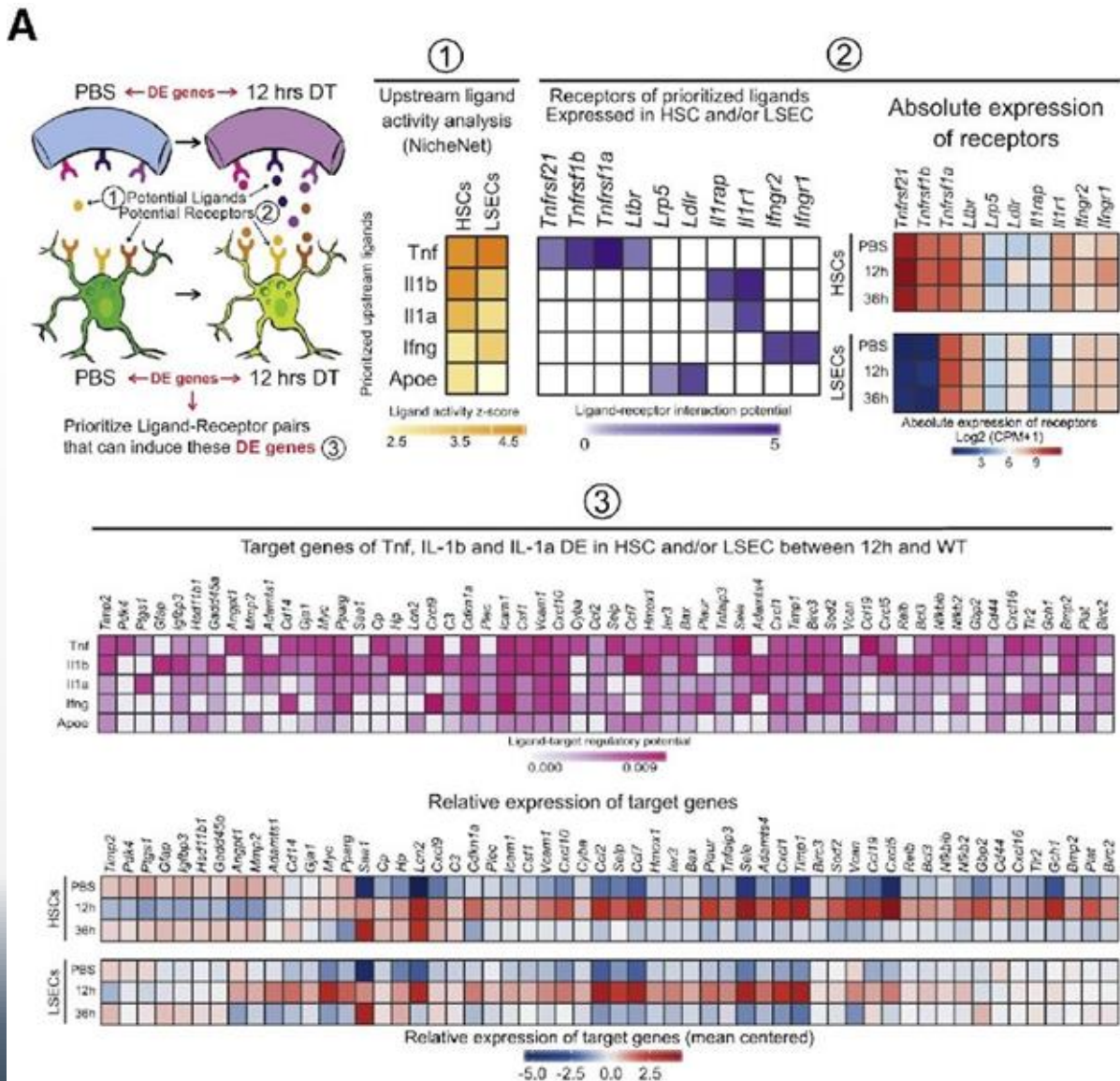
Stellate Cells, Hepatocytes, and Endothelial Cells Imprint the Kupffer Cell Identity on Monocytes Colonizing the Liver Macrophage Niche

Johnny Bonnardel^{1, 2, 16}, Wouter T'Jonck^{1, 2, 16}, Djoere Gaubomme^{3, 4}, Robin Browaeys^{5, 6}, Charlotte L. Scott^{1, 2, 7}, Liesbet Martens^{2, 5}, Bavo Vanneste^{1, 2}, Sofie De Prijck^{1, 2}, Sergei A. Nedospasov^{8, 9}, Anna Kremer^{2, 10}, Evelien Van Hamme^{2, 10}, Peter Borghgraef^{2, 10}, Wendy Toussaint^{4, 11}, Pieter De Bleser^{2, 5}, Inge Mannaerts¹², Alain Beschin^{13, 14}, Leo A. van Grunsven¹², Bart N. Lambrecht^{4, 11}, Tom Taghon¹⁵, Saskia Lippens^{2, 10}, Dirk Elewaut^{3, 4}, Yvan Saeys^{5, 6, 17}, Martin Guilliams^{1, 2, 17, 18}

What intercellular signaling drives
activation of Hepatic Stellate cells?

Predicts ligands (TNF) with more active
targets in receiving cell types; verified by
blocking in vivo

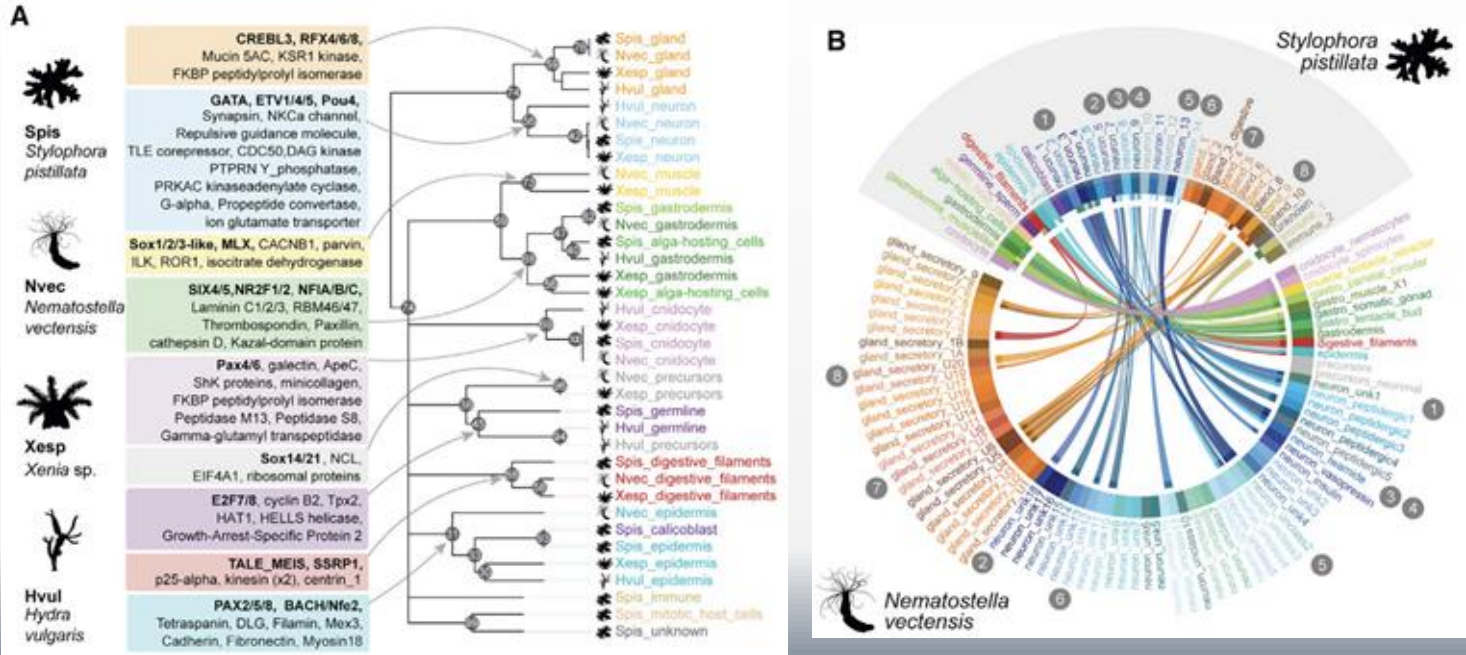
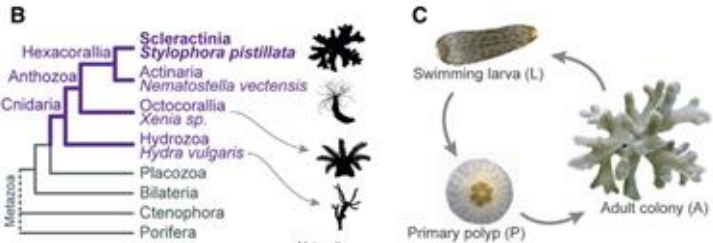
Other tools focus on identifying GRN or
CCN specifically



Uses of scRNA-seq – evolutionary comparisons

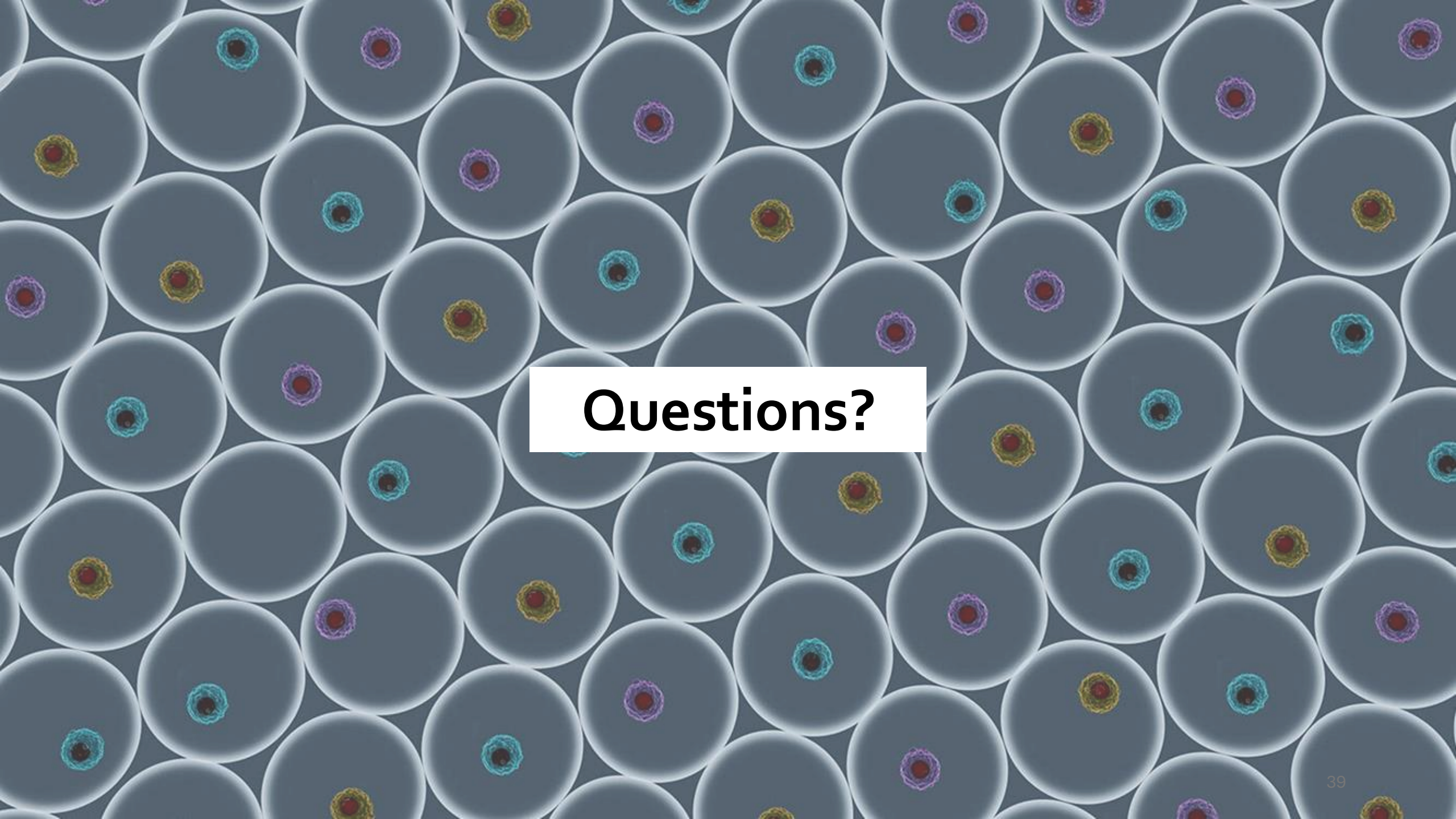
A stony coral cell atlas illuminates the molecular and cellular basis of coral symbiosis, calcification, and immunity

Shani Levy ⁶ • Anamaria Elek ⁶ • Xavier Grau-Bové • ... Amos Tanay • Tali Mass ⁶   • Arnau Sebé-Pedrós ⁷   • Show all authors • Show footnotes



Comparison of Cnidaria cell type lineages

Finds origins of immune and calcification gene programmes

The background of the slide is a repeating pattern of light blue circles, each containing a smaller, darker blue circle in the center. These central circles are surrounded by a thin, irregular ring of color, which can be red, yellow, or green. The overall effect is a dense, textured field of these colored circles.

Questions?